University of Antwerp
Algorithms and Complexity
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Exercises session 7

Reductions

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1 Definitions

Definition 1 (Graph colorability) *A graph $G = (V, E)$ is k-colorable if there's an assignment of k colours to vertices $c : V \to 1, \dots, k$ such that for every $(u, v) \in E$, $c(u) \neq c(v)$*

Definition 2 (Planar graph) *A planar graph is a graph which* ***can*** *be drawn in the plane without any edges crossing*

2 Session exercises

Prove that the following problems are NP-complete

1. **3-COLORING**. Given $G = (V, E)$ decide if the graph is 3-colorable.
2. **PLANAR 3-COLORING**. Given $G = (V, E)$ decide if G is a planar 3-colorable graph.
3. **SET COVER**. Lets $\mathcal{U}$ be a set of elements, $\mathcal{S}$ a collection of subsets of $\mathcal{U}$ and k an integer. Decide if there's a collection of at most k subsets whose union is $\mathcal{U}$.
4. **KNAPSACK.** Given n items of sizes $s_1, \dots, s_n$ and values $v_1, \dots, v_n$, a capacity B and a value V, is there any subset $S \subseteq \{1, \dots, n\}$ such that $\sum_{i \in S} s_i \leq B$ and $\sum_{i \in S} v_i \geq V$

① 3-COLORING IS NP-COMPLETE

- **3-COLORING IS IN NP**

We will prove this via a polinomial time verifier.

Our verifier does the following:

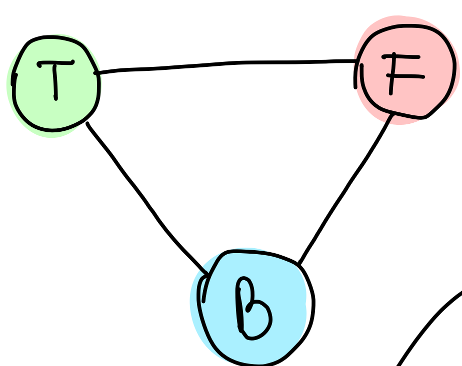
- take graph $G=(V,E)$ and a colouring c
- check if c is a proper colouring
  - ↳ how long does this take?
    - $O(|V|^2)$ / $O(|E|)$.
      - ↳ for every edge in E, check if they have different colours.

• 3-COLORING IS NP-HARD

We will do a reduction from 3-SAT, this is, given an instance φ for 3-SAT, we will construct G_φ such that

$$\underbrace{\varphi \in 3\text{-SAT}}_{\text{3-SAT instance}} \iff \underbrace{G_\varphi \in 3\text{-COL}}_{\text{3-COL instance.}}$$

So, let φ be a CNF formula with clauses $C_1, \dots, C_m$ and variables $x_1, \dots, x_n$.



create a triangle

true
false
base

$\Big\}$ the colors

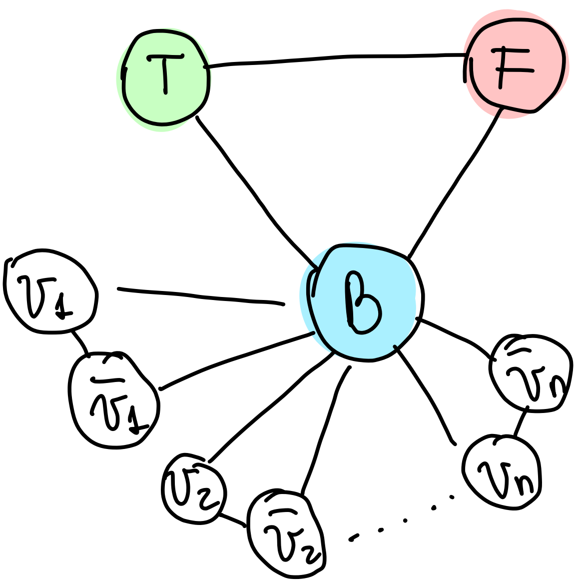
this triangle will be a part of our graph G_φ

↳ therefore G_φ already need at least 3 colors.

now, for every literal x_i we will create 2 vertices

$$x_i \rightarrow v_i$$
$$\rightarrow \bar{v}_i$$

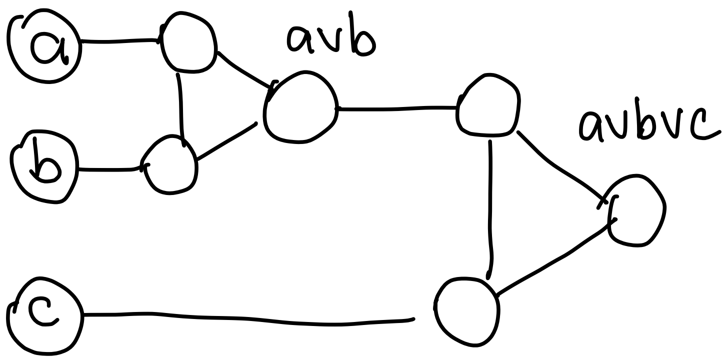
and we will create edges (v_i, B) $(v_1, \bar{v}_1)$ $(\bar{v}_1, B)$



this forces that either v_i or $\bar{v}_i$ will get color T

↳ so we are inducing a truth assignment!

now, for every clause $c_i = (a \vee b \vee c)$ we will create a small gadget graph



think of this as a CIRCUIT.

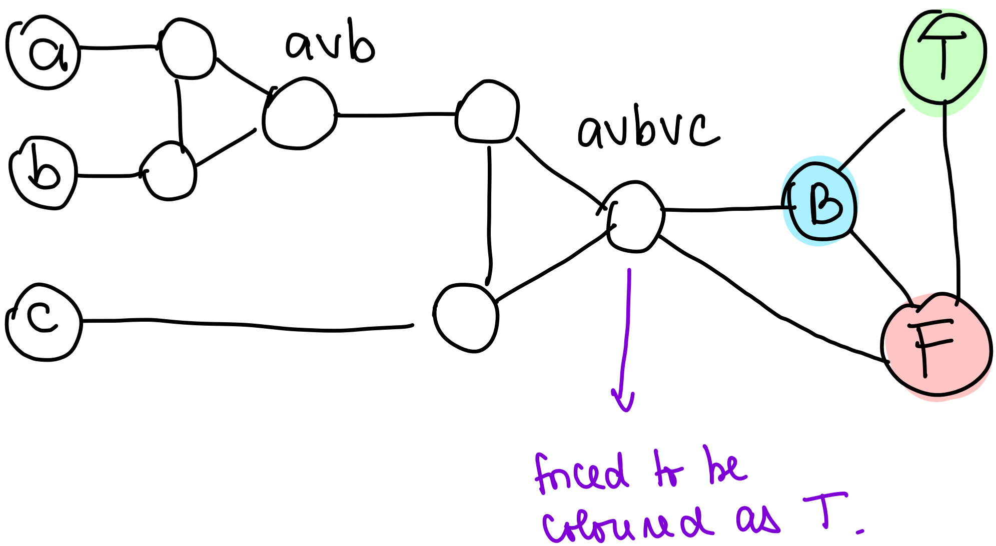
OBSERVATION

the gadget satisfies the following properties

1. if a, b, c are all coloured to F, then the output will be F

2. if any a, b, c is T, then there exists a 3-colouring where the output is coloured as T.

now, for each clause we connect the gadget with our triangle



now, we need to prove

$$\varphi \text{ is satisfiable} \iff G_\varphi \text{ is 3-colorable}$$

$\Rightarrow$) suppose φ is satisfiable

let $\tau(x_1), \dots, \tau(x_n)$ be a satisfying assignment for φ.

- if $\tau(x_i) = \text{TRUE}$, then we color $c(v_i) = T$, therefore $c(\bar{v}_i) = F$
- as φ is satisfiable, for every clause $C_j = (a \vee b \vee c)$ at least $a, b,$ or c must be true, so the gadget can be 3-coloured

$\therefore$ G_φ is 3-COLORABLE

($\Leftarrow$) suppose G_φ is 3-COLORABLE.

As G_φ is 3-COLORABLE, then it has a valid colouring.

We will build a truth assignment for φ out of this colouring

$$\sigma(x_i) = \text{TRUE} \quad \text{if} \quad c(v_i) = T$$

$$\sigma(x_i) = \text{FALSE} \quad \text{if} \quad c(v_i) = F.$$

Now, by contradiction, we will assume that this is not a satisfying assignment, so, there's at least one clause $(a \vee b \vee c)$ that is not satisfied, this is $a = b = c = \text{FALSE}$.

<u>But</u>... if this is the case, then the gadget output will be colored as F, but then the colouring wouldn't be a valid colouring $\times$

② PLANAR 3-COLOURING IS NP-COMPLETE.

• PLANAR 3-COL IS IN NP.

Same as the verifier for 3-COL

• PLANAR 3-COL IS NP-HARD

We reduce from 3-COL

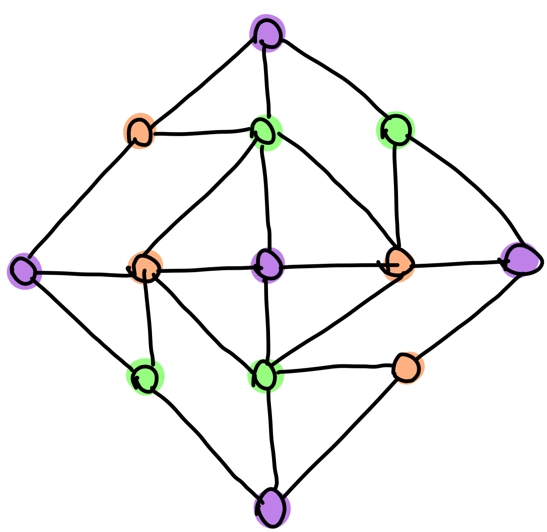
$$G \in \text{3-COL} \iff \tilde{G} \in \text{PLANAR 3-COL}$$

So, we will build $\tilde{G}$ as follows,

- $\tilde{G}$ has to be planar, so we have to replace the edge crossings

↓

we will use the following gadget.



Properties of this gadget

(1) every valid 3-coloring has opposite corners of the same color

(2) any colouring of the corners extends to a colouring of the entire gadget.

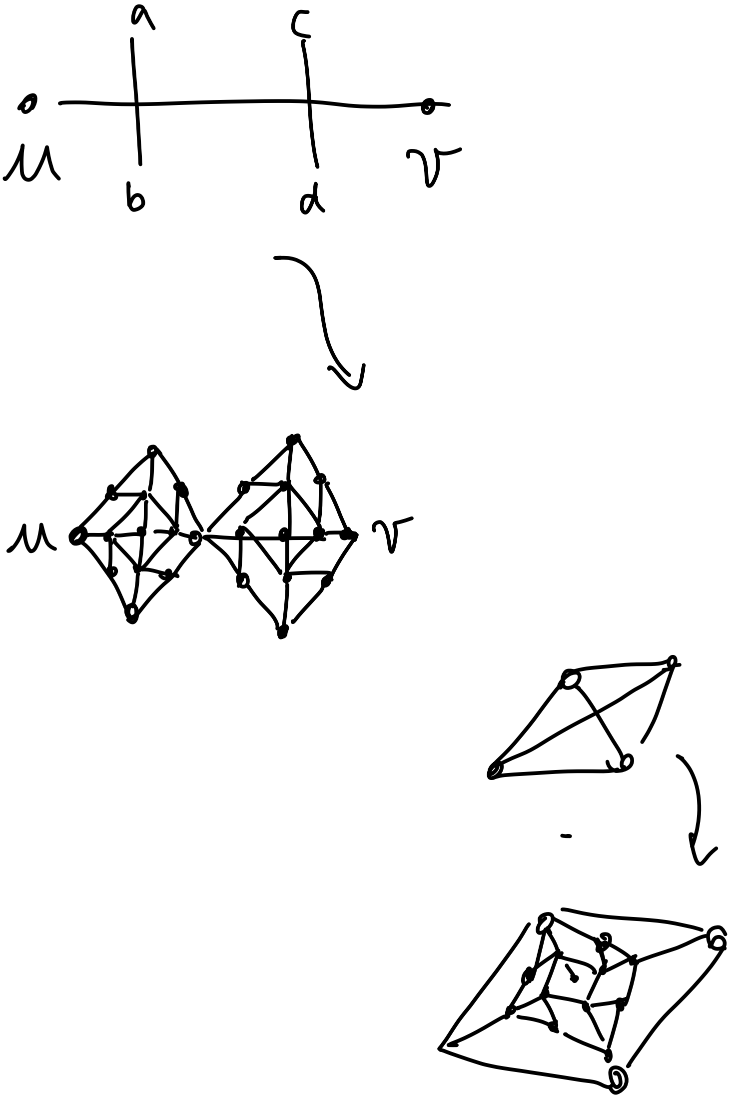
If an edge in G is crossed by multiple edges, the gadgets that replace those crossings need to be linked together at the edges.
(to ensure that the nodes at either end must be different colours).

TO PROVE:

- $\tilde{G}$ is planar
- $\tilde{G}$ is 3-COL $\iff$ G is 3-COL.

Hint: the colouring of G is extended to $\tilde{G}$ because of the gadget.

Example



③ SET COVER IS NP-COMPLETE.

- SET COVER IS IN NP

We also do this via verifier: we are given a certificate

↳ collection of subsets $S_{i_1}, \dots, S_{i_k}$

We only have to check

$$\bigcup_{j=1}^{k} S_{i_j} = \mathcal{U}$$ } can be done in $O(|\mathcal{U}|)$.

- SET COVER IS NP-HARD

We will reduce from VERTEX-COVER

> VERTEX COVER
> INPUT: $G=(V,E)$, integer k
> Is there a set of at most k vertices $\mathcal{V}$ such that $\forall (u,v) \in E$, $u \in \mathcal{V}$ or $v \in \mathcal{V}$.

We want

$\langle G, k \rangle \in$ VERTEX-COVER

$\iff \langle \bar{\mathcal{U}}, \bar{S}, k \rangle \in$ SET-COVER.

We build the instance for set cover as follows

- $\mathcal{U} = E$ (set of edges) → vertices (?)
- S_u $\forall u \in V$ $S_u = \{v : (u,v) \in E\}$
- same k as in vertex-cover

claim:

(1) if k sets $S_{u_1}, \dots, S_{u_k}$ cover $\mathcal{U}$, then every edge is adjacent to at least one vertex from $u_1, \dots, u_k$. $\therefore$ $u_1, \dots, u_k$ is a vertex cover

(2) if $u_1, \dots, u_k$ form a vertex cover, then S_{u_i} cover edges adjacent to u_i. therefore $S_{u_1}, \dots, S_{u_k}$ form a set cover of $\mathcal{U}$

Proof extended

- $\bar{\mathcal{U}} = V$
- $\bar{S} = \bigcup_{u \in V} S_u$ $\qquad S_u = \{v : (u,v) \in E\}$
- some k

claim:

$\langle \bar{\mathcal{U}}, \bar{S}, k \rangle \in S\text{-}C \iff \langle G, k \rangle \in$ Vertex-Cover

$\Rightarrow$ assume

$\langle \bar{\mathcal{U}}, \bar{S}, k \rangle \in S\text{-}C$

$\Rightarrow \exists u_1, \dots, u_k$ s.t

$\bigcup_{i=1}^{k} S_{u_i} = \bar{\mathcal{U}}$ $\quad \therefore \forall v \in V \;\; \exists i \in [1,k]$ s.t $(u_i, v) \in E$

$\therefore u_1, \dots, u_k$ is a vertex cover

$\Leftarrow$ assume

$\langle G, k \rangle \in VC$

$\Rightarrow \exists u_1, \dots, u_k$ s.t $\forall v \in V.\ \exists i \in [1,k]$ s.t $(u_i, v) \in E$

$\therefore \forall v \in V = \bar{\mathcal{U}}$, $\exists i \in [1,k]$ s.t $v \in S_{u_i}$

$\therefore S_{u_1}, \dots, S_{u_k}$ is a set cover for $\bar{\mathcal{U}}$

④ KNAPSACK IS NP-COMPLETE

• KNAPSACK IS IN NP

as a certificate, we are given the set of items $S \subseteq \{1, \dots, n\}$ and we need to verify

$$\left.\begin{array}{l} \bullet \ \sum_{i \in S} s_i \leq B \\ \bullet \ \sum_{i \in S} v_i \geq V \end{array}\right\} \text{linear time } \Theta(n)$$

• KNAPSACK IS NP-HARD

we will reduce from PARTITION

PARTITION

INPUT: a set of integers $\mathcal{A} = a_1, \dots, a_n$

$\mathcal{A} \in$ PARTITION if there exists $A, B \subseteq \mathcal{A}$, $A \cup B = \mathcal{A}$, $A \cap B = \emptyset$ and $\sum_{a \in A} a = \sum_{b \in B} b$.

then we build the instance for KNAPSACK as follows

$$s_i = a_i \qquad \forall i = 1, \dots, n \qquad B = V = \frac{\sum_{i=1}^{n} a_i}{2}$$
$$v_i = a_i$$

now, lets see if the reduction is correct.

$a_1, \dots, a_n \in$ PARTITION $\iff (s_1, \dots, s_n), (v_1, \dots, v_n),$
$B, V \in$ KNAPSACK

$\Rightarrow$) assume that $a_1, \dots, a_n \in$ PARTITION

then there exists S, T s.t

$$\sum_{i \in T} a_i = \sum_{i \in S} a_i = \frac{\sum_{i=1}^{n} a_i}{2}$$

then our KNAPSACK can contain elements in S, and we check that

$$\sum_{i \in S} a_i = B \qquad \sum_{i \in S} a_i = \sum_{i \in S} v_i = \underbrace{V}_{\frac{1}{2}\sum_{i=1}^{n} a_i} \; ; \; \sum_{i \in S} s_i = \sum_{i \in S} a_i = \underbrace{B}_{\frac{1}{2}\sum_{i=1}^{n} a_i} .$$

$(\Leftarrow)$ assume that $(v_1, \ldots, v_n)$
$(s_1, \ldots, s_n)$

Consider S as the chosen set of items, and

$$T = \mathcal{U} - S$$

We have that

$$\sum_{i \in S} s_i = \sum_{i \in S} a_i \leq B = \frac{\sum_{i=1}^{n} a_i}{2}$$

and

$$\sum_{i \in S} v_i = \sum_{i \in S} a_i \geq V = \frac{\sum_{i=1}^{n} a_i}{2}$$

$$\Rightarrow \sum_{i \in S} a_i = \frac{1}{2} \sum_{i=1}^{n} a_i$$

$$\sum_{i \in T} a_i = \frac{1}{2} \sum_{i \in T}^{n} a_i$$

$$\therefore \sum_{i \in S} a_i = \sum_{i \in T} a_i$$

$\blacksquare$